

Discrete Time:

First-Order

Difference Equations

Discrete Variable:

When time is instead, taken to be a discrete variable, so that the variable t is allowed to make integer values only, the concept of the derivative obviously will no longer be appropriate.

Variable y :

The pattern of change of the variable y must be described by so-called difference, rather than by derivatives or differentials of $y(i)$.

Difference Equations:

The technique of differential

equations will give way to those of difference equations.

Discrete Time, Differences, and Difference Equations:

$$\Delta y_t \equiv y_{t+1} - y_t \quad \text{--- (i)}$$

$$\Delta y_t = 2 \quad \text{--- (ii)}$$

By virtue of eq. (i), we can rewrite eq. (ii) as,

$$y_{t+1} - y_t = 2$$

$$y_{t+1} = y_t + 2$$

$$y_t - y_{t-1} = 2$$

we may express it equivalently

as:

$$y_{t+2} - y_{t+1} = 2$$

Solving a First-Order Difference Equations

Iterative Method:

Before developing a general method of attack, let us first explain a relatively pedestrian method, the iterative method — which, though crude, will prove immensely revealing of the essential nature of a so-called solution.

Example:

Find the solution of the difference equation, assuming an initial value of $y_0 = 15$. To carry out the iterative process, it is more convenient to use the alternative form of the difference equation, namely, $y_{t+1} = y_t + 2$, with $y_0 = 15$.

We can deduce step-by-step that

$$y_{t+1} = y_t + 2$$

$$y_0 = 15$$

$$t = 0$$

$$\begin{aligned} y_1 &= y_0 + 2 \\ &= 15 + 2 \\ &= \boxed{17} \end{aligned}$$

$$t = 1$$

$$\begin{aligned} y_{1+1} &= y_1 + 2 \\ y_2 &= (y_0 + 2) + 2 \\ &= y_0 + 2(2) \\ &= 15 + 2(2) \\ &= 15 + 2 + 2 \\ &= \boxed{19} \end{aligned}$$

$$t = 2$$

$$\begin{aligned} y_{2+1} &= y_2 + 2 \\ y_3 &= ((y_0 + 2) + 2) + 2 \\ &= 19 + 2 \\ &= \boxed{21} \end{aligned}$$

and, in general, for any period t ,

$$y_t = y_0 + t(2) = 15 + 2t \quad \text{--- eq.}$$

Example,

Solve the difference equation this time, let the initial value be unspecified and denoted simply by y_0 . Again it is more convenient to work with the alternative version in, namely, $y_{t+1} = 0.9y_t$.

By iteration, we have

$$y_{t+1} = 0.9y_t$$

$$t = 0$$

$$y_{0+1} = 0.9y_0$$

$$y_1 = \boxed{0.9y_0}$$

$$t = 1$$

$$y_{1+1} = 0.9y_1$$

$$y_2 = 0.9(0.9y_0)$$

$$y_2 = \boxed{(0.9)^2 y_0}$$

$$t = 2$$

$$y_{2+1} = 0.9y_2$$

$$y_3 = 0.9(0.9)^2 y_0$$

$$y^3 = \boxed{(0.9)^3 y_0}$$

These can be summarized into
the solution

$$y_t = (0.9)^t y_0$$

Example,

Solve the homogeneous difference equation

$$my_{t+1} - ny_t = 0$$

Upon normalizing and transposing,
this may be written as:

$$y_{t+1} = \left(\frac{n}{m}\right) y_t$$

$$y_t = \left(\frac{n}{m}\right)^t y_0$$

The solution of the general
homogeneous difference equation

$$y_t = Ab^t$$

General Method:

The solution to the first-order difference equation

$$y_{t+1} + ay_t = c$$

where a and c are two constants.

The general solution of the reduced equation

$$y_{t+1} + ay_t = 0$$

If these values of y_t and y_{t+1} hold, the homogenous equation will become

$$y_t = Ab^t$$

$$y_{t+1} = Ab^{t+1}$$

$$y_{t+1} + ay_t = 0$$

$$Ab^{t+1} + aAb^t = 0$$

which, upon canceling the nonzero common factor Ab^t , yields

$$b + a = 0, \quad b = -a$$

The complementary function should be written as:

$$y_c = (A b^t) = A(-a)^t$$

The particular integral can be written as:

$$y_p = \frac{c}{1+a} \quad (a \neq -1)$$

Adding y_c and y_p together, we may now write the general solution in one of the two following forms:

$$y_t = A(-a)^t + \frac{c}{1+a}$$

[general solution, case of $a \neq -1$]

$$y_t = A(-a)^t + 0$$

[general solution, case of $a = -1$]

find A ?
when $t=0$

$$y_0 = A(-a)^0 + \frac{c}{1+a}$$

$$y_0 = A + \frac{c}{1+a}$$

$$A = y_0 - \frac{c}{1+a}$$

$$y_c = A(-a)^t \\ = \left(y_0 - \frac{c}{1+a} \right) (-a)^t$$

$$y_p = \frac{c}{1+a}$$

General solution:

$$y_t = y_c + y_p$$

$$= \left(y_0 - \frac{c}{1+a} \right) (-a)^t + \frac{c}{1+a}$$

[definite solution, case of $a \neq -1$]

Example

Solve the first-order difference equation:

$$y_{t+1} - 5y_t = 1$$

$$y_0 = 7/4$$

$$\alpha = -5, c = 1$$

$$y_p = \frac{c}{1+\alpha} = \frac{1}{1+(-5)} = \frac{1}{-4}$$

$$\boxed{y_p = \frac{1}{-4}}$$

$$\begin{aligned} y_c &= A(-\alpha)^t \\ &= A(-(-5))^t \end{aligned}$$

$$\boxed{y_c = A(5)^t}$$

General Solution:

$$x_t = y_c + y_p$$

$$\boxed{y_t = A(5)^t + \frac{1}{-4}}$$

find $A = ?$

$$A = \left(y_0 - \frac{c}{1+a} \right)$$

$$A = \left(\frac{7}{4} - \frac{1}{1+(-5)} \right)$$

$$A = \left(\frac{7}{4} - \frac{1}{-4} \right)$$

$$A = \left(\frac{7+1}{4} \right)$$

$$A = \left(\frac{8}{4} \right)$$

$$\boxed{A = 2}$$

General Solution: put in value A

$$y_t = y_c + y_p$$

$$\boxed{y_t = 2(5)^t + \frac{1}{-4}}$$

Ans

Exercise 17.2

Q:1

Convert the following
difference equation into
the form of (17.2").

(a) $\Delta y_t = 7$

$$\Delta y_t = 7$$

$$y_{t+1} - y_t = 7$$

$$y_{t+1} = 7 + y_t$$

let,

$$t = 0$$

$$y_{0+1} = 7 + y_0$$

$$y_1 = 7 + y_0$$

$$t = 1$$

$$y_{1+1} = 7 + y_1$$

$$y_2 = 7 + (7 + y_0)$$

$$y_2 = 14 + y_0$$

$$t = 2$$

$$y_{2+1} = 7 + y_2$$

$$y_3 = 7 + (14 + y_0)$$

$$y_3 = 21 + y_0$$

$$t=3$$

$$y_{3+1} = 7 + y_3$$

$$y_4 = 7 + (21 + y_0)$$

$$y_4 = 28 + y_0$$

$$(b) \Delta y_t = 0.3y_t$$

$$y_{t+1} - y_t = 0.3y_t$$

$$y_{t+1} = 0.3y_t + y_t$$

Let

$$t=0$$

$$y_{0+1} = 0.3y_0 + y_0$$

$$y_1 = 0.3y_0 + y_0$$

$$t=1$$

$$y_{1+1} = 0.3y_1 + y_1$$

$$y_2 =$$

$$\underline{(c)} \Delta y_t = 2y_t - 9$$

$$\Delta t+1 - y_t = 2y_t - 9$$

$$y_{t+1} - y_t = 2y_t - 9$$

$$y_{t+1} = 2y_t - 9 + y_t$$

let,

$$t=0$$

$$y_{0+1} = 2y_0 - 9 + y_0$$

$$y_1 = 2y_0 - 9 + y_0$$

$$y_1 = 3y_0 - 9$$

$$t=1$$

$$y_{1+1} = 2y_1 - 9 + y_1$$

$$y_2 = 3y_1 - 9$$

$$y_2 = 3(3y_0 - 9) - 9$$

$$y_2 = 9y_0 - 27 - 9$$

$$y_2 = 9y_0 - 36$$

$$t=2$$

$$y_{2+1} = 2y_2 - 9 + y_2$$

$$y_3 = 3y_2 - 9$$

$$y_3 = 3(9y_0 - 36) - 9$$

$$y_3 = 27y_0 - 108 - 9$$

$$y_3 = 27y_0 - 117$$

$$t=3$$

$$y_{3+1} = 2y_3 - 9 + y_3$$

$$y_4 = 3y_3 - 9$$

$$y_4 = 3(27y_0 - 117) - 9$$

$$y_4 = 81y_0 - 351 - 9$$

$$y_4 = 81y_0 - 360$$

Q:2

Solve the following difference equations by iteration.

(a) $y_{t+1} = y_t + 1$ ($y_0 = 10$)

$$y_{t+1} = y_t + 1$$

Let,

$$t = 0$$

$$y_{0+1} = y_0 + 1$$

$$y_1 = 10 + 1$$

$$y_1 = 11$$

$$t = 1$$

$$y_{1+1} = y_1 + 1$$

$$y_2 = 11 + 1$$

$$y_2 = 12$$

$$t = 2$$

$$y_{2+1} = y_2 + 1$$

$$y_3 = 12 + 1$$

$$y_3 = 13$$

$$t=3$$

$$y_{3+1} = y_{3+1}$$

$$y_4 = 13+1$$

$$y_4 = 14$$

$$(b) \ y_{t+1} = a y_t \quad (y_0 = \beta)$$

$$y_{t+1} = a y_t$$

Let:

$$t=0$$

$$y_{0+1} = a y_0$$

$$y_1 = a\beta$$

$$t=1$$

$$y_{1+1} = a y_1$$

$$y_2 = a(a\beta)$$

$$y_2 = a^2\beta$$

$$t=2$$

$$y_{2+1} = a y_2$$

$$y_3 = a(a^2\beta)$$

$$y_3 = a^3\beta$$



$$t=3$$

$$y_{3+1} = ay_3$$

$$y_4 = a(a^3\beta)$$

$$y_4 = a^4\beta$$

$$\textcircled{c} \quad y_{t+1} = ay_t - \beta \quad (y_t = y_0 \text{ when } t=0)$$

$$y_{t+1} = ay_t - \beta$$

let

$$t=0$$

$$y_{0+1} = ay_0 - \beta$$

$$y_1 = ay_0 - \beta$$

$$t=1$$

$$y_{1+1} = ay_1 - \beta$$

$$y_2 = a(ay_0 - \beta) - \beta$$

$$y_2 = a^2y_0 - a\beta - \beta$$

$$t=2$$

$$y_{2+1} = ay_2 - \beta$$

$$y_3 = a(a^2y_0 - a\beta - \beta) - \beta$$

$$y_3 = a^3y_0 - a^2\beta - a\beta - \beta$$

$$t=3$$

$$y_{3+1} = ay_3 - \beta$$

$$y_4 = a(a^3 y_0 - a^2 \beta - a\beta - \beta) - \beta$$

$$y_4 = a^4 y_0 - a^3 \beta - a^2 \beta - a\beta - \beta$$

Q:4

For each of the following
difference equations, use the
procedure illustrated in the
derivation of (17.8) and (17.9).

To find y_c , y_p , and the
definite solution:

$$(a) \quad y_{t+1} + 3y_t = 4 \quad (y_0 = 4)$$

$$a = 3, c = 4, y_0 = 4$$

$$y_p = \frac{c}{1+a} = \frac{4}{1+3} = \frac{4}{4} = \underline{1}$$

$$y_c = A(-a)^t$$

$$y_c = \left(y_0 - \frac{c}{1+a} \right) (-a)^t$$

$$y_c = \left(4 - \frac{4}{1+3} \right) (-3)^t$$

$$y_c = \left(4 - \frac{4}{4} \right) (-3)^t$$

$$y_c = (4-1) (-3)^t$$

$$\boxed{y_c = (3) (-3)^t}$$

General Solution:

$$y_t = y_c + y_p$$

$$y_t = 3(-3)^t + 1 \quad \underline{\text{Ans}}$$

(b) $2y_t + 1 - y_t = 6 \quad (y_0 = 7)$

$$a = -1, c = 6, y_0 = 7$$

$$y_p = \frac{c}{1+a} = \frac{6}{1+(-1)} = \boxed{6}$$

$$y_c = A(-a)^t$$

$$y_c = \left(y_0 - \frac{c}{1+a} \right) (-a)^t$$

$$y_c = (7 - 6) (-1)^t$$

$$\boxed{y_c = 1(-1)^t}$$

General Solution:

$$y_t = y_c + y_p$$

$$y_t = 1(-1)^t + 6$$

Ans

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(c) $y_{t+1} = 0.2y_t + 4$ ($y_0 = 4$)

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$$y_{t+1} - 0.2y_t = 4$$

$$a = -0.2, c = 4, y_0 = 4$$

$$y_p = \frac{c}{1+a} = \frac{4}{1+(-0.2)} = \frac{4}{1/5} = \boxed{5}$$

$$y_c = A(-a)^t$$

$$y_c = \left(y_0 - \frac{c}{1+a} \right) (-a)^t$$

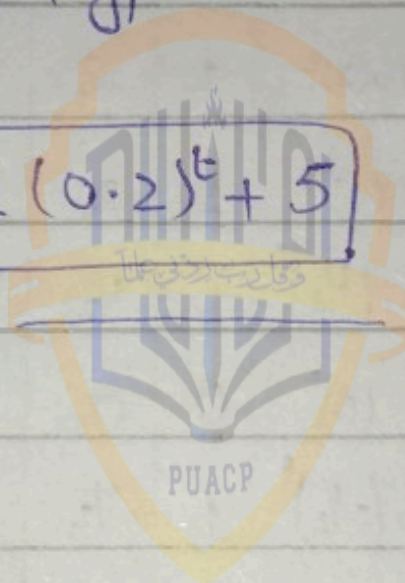
$$y_c = (4-5)(-(-0.2))^t$$

$$\boxed{y_c = -1(0.2)^t}$$

General Solution:

$$y_t = y_c + y_p$$

$$\boxed{y_t = -1(0.2)^t + 5} \quad \underline{\text{ans}}$$



The Dynamic Stability of Equilibrium

Table:

A classification of the value of b .

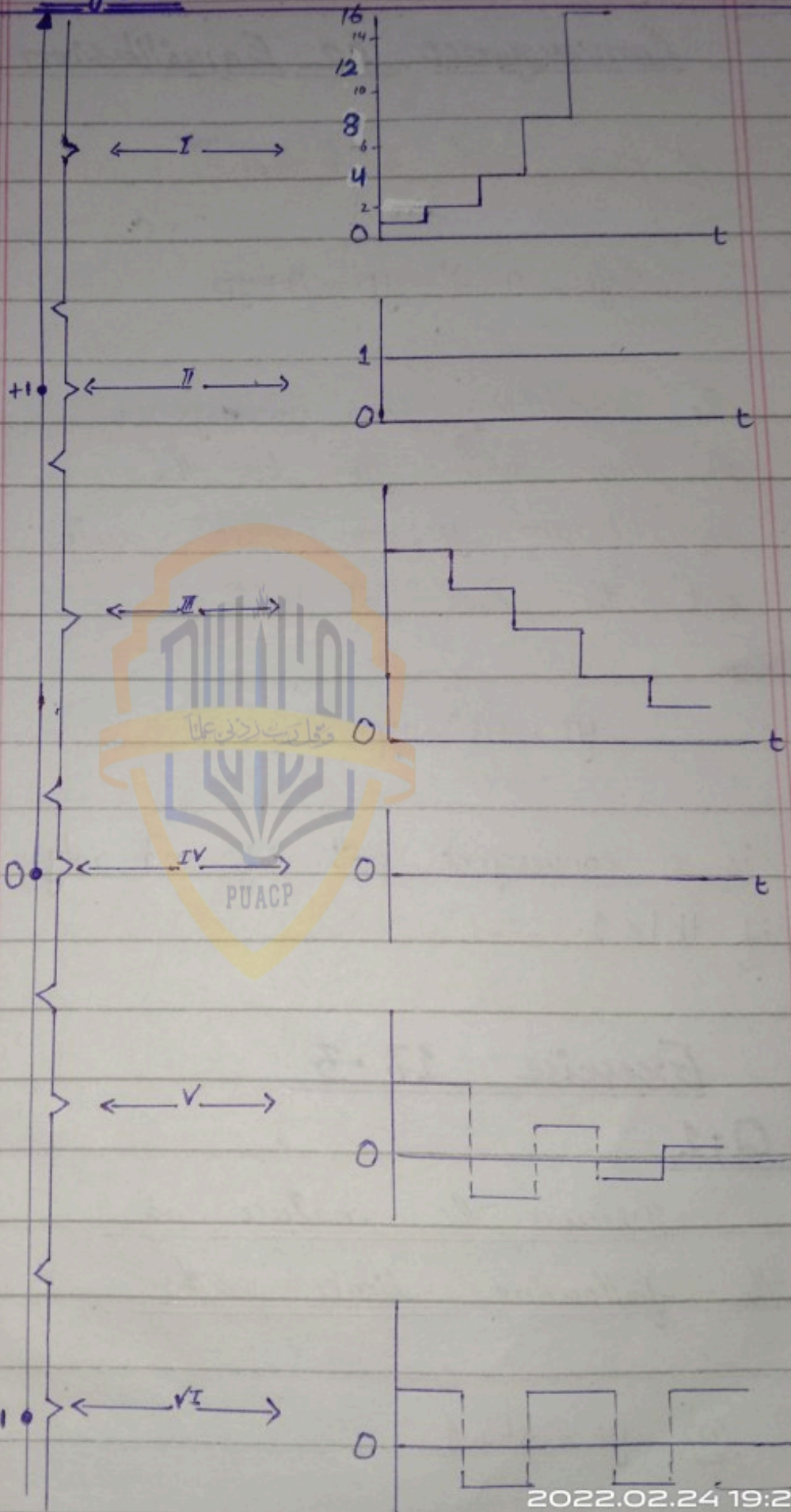
Re- gion	Value of b		Value of b^t	Value of b^t in Different Time period				
				$t=0$	$t=1$	$t=2$	$t=3$	$t=4$
I	$b > 1$	$(b > 1)$	e.g., $(2)^t$	1	2	4	8	16
II	$b = 1$	$(b = 1)$	$(1)^t$	1	1	1	1	1
III	$0 < b < 1$	$(b < 1)$	e.g., $(\frac{1}{2})^t$	1	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$
IV	$b = 0$	$(b = 1)$	$(0)^t$	0	0	0	0	0
V	$-1 < b < 0$	$(b < 1)$	e.g., $(-\frac{1}{2})^t$	1	$-\frac{1}{2}$	$\frac{1}{4}$	$-\frac{1}{8}$	$\frac{1}{16}$
VI	$b = -1$	$(b = 1)$	$(-1)^t$	1	-1	1	-1	1
VII	$b < -1$	$(b > 1)$	e.g., $(-2)^t$	1	-2	4	-8	16

The essence of the proceeding discussion can be conveyed in the following general statement:

The time path of b^t ($b \neq 0$) will be

Nonoscillatory } if $\begin{cases} b > 0 \\ b < 0 \end{cases}$
 Oscillatory }
 Divergent } if $\begin{cases} |b| > 1 \\ |b| < 1 \end{cases}$
 Convergent }

Figure:



Convergence to Equilibrium

A time path such as:

$$y_t = A(1)^t + y_p = A + y_p$$

The condition for convergence of time path y_t to the equilibrium y_p , we must rule out the case of $b=1$

$$y_t = Ab^t + y_p$$

is a convergent path if and only if $|b| < 1$.